SIMATS Engineering • CSE • CSA0802 — Python Programming • 26-08-2026 • Category: Class Work (22.8.26) • Student Submissions Report

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Q1. Compute the sum of all natural numbers

CODE

```
total = 0
for n in range(1, 500):
    if n % 3 == 0 or n % 5 == 0:
        total += n

print(total)
```

OUTPUT

```
57918
```

Q2. Calculate natural numbers.

CODE

```
sum1 = 0
sum2 = 0

for i in range(1, 201):
    sum1 = sum1 + i
    sum2 = sum2 + i * i

answer = sum1 * sum1 - sum2

print(answer)
```

OUTPUT

```
401323300
```

Q3. Triangle pyramid

CODE

```
n = int(input())
a = [list(map(int, input().split())) for _ in range(n)]

for i in range(n - 2, -1, -1):
    for j in range(i + 1):
        a[i][j] += max(a[i + 1][j], a[i + 1][j + 1])

print("Maximum sum =", a[0][0])
```

INPUT

```
4
3
4 6
2 7 6
8 5 9 3
```

OUTPUT

```
Maximum sum = 25
```

Q4. Finds the length of the longest sequence of consecutive numbers

CODE

```
a=list(map(int,input().replace('[','').replace(']','').replace(',',' ').split()))
s=set(a);ans=0
for x in s:
    if x-1 not in s:
        y=x
        while y in s:y+=1
        ans=max(ans,y-x)
print(ans)
```

INPUT

```
[100, 4, 200, 1, 3, 2]
```

OUTPUT

```
4
```

Q5. 2D matrix

CODE

```
matrix = [
    [1, 2, 3],
    [4, 0, 6],
    [7, 8, 9]
]

rows = set()
cols = set()

for i in range(len(matrix)):
    for j in range(len(matrix[0])):
        if matrix[i][j] == 0:
            rows.add(i)
            cols.add(j)

for i in range(len(matrix)):
    for j in range(len(matrix[0])):
        if i in rows or j in cols:
            matrix[i][j] = 0

print(matrix)
```

OUTPUT

```
[[1, 0, 3], [0, 0, 0], [7, 0, 9]]
```